

PORTFOLIO PROJECT P08

AI Project Manager

Trello → n8n → deterministic risk rules → Gemini founder report → Telegram.

- OVERDUE
- BLOCKED
- AT RISK
- ON TRACK

Founders find out about late projects too late

Project data is on a Trello board, but nobody reads twelve cards every morning. Overdue work, blockers, hours running over and stale cards stay hidden until a client asks.

Manual status checks

Someone has to open every card and compare dates and hours.

Hidden dependencies

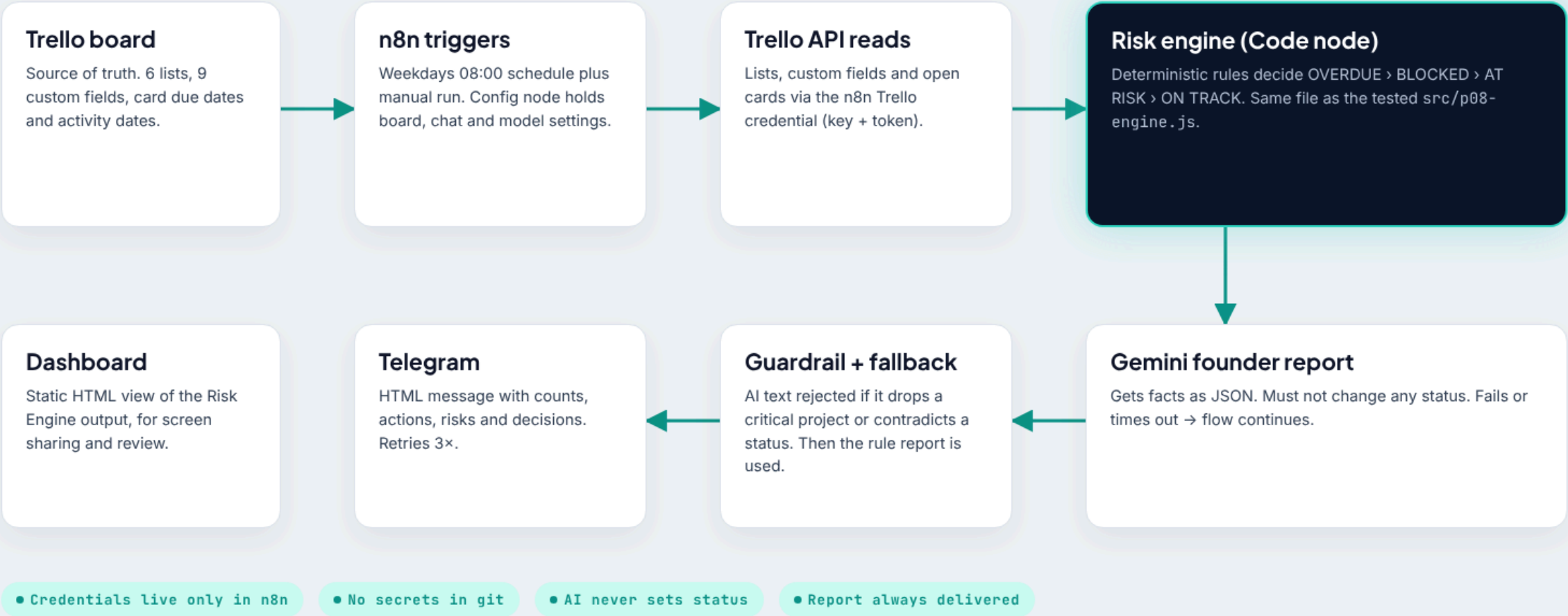
A blocked card quietly delays the cards that depend on it.

AI alone is not reliable

An LLM asked "is this project at risk?" can give a different answer each time.

Trello → n8n → rules → Gemini → Telegram

Each layer does one job. Trello holds the data, n8n runs the flow, the rule engine decides health, Gemini only writes the summary, and Telegram delivers it.



Health is decided by rules, not by AI

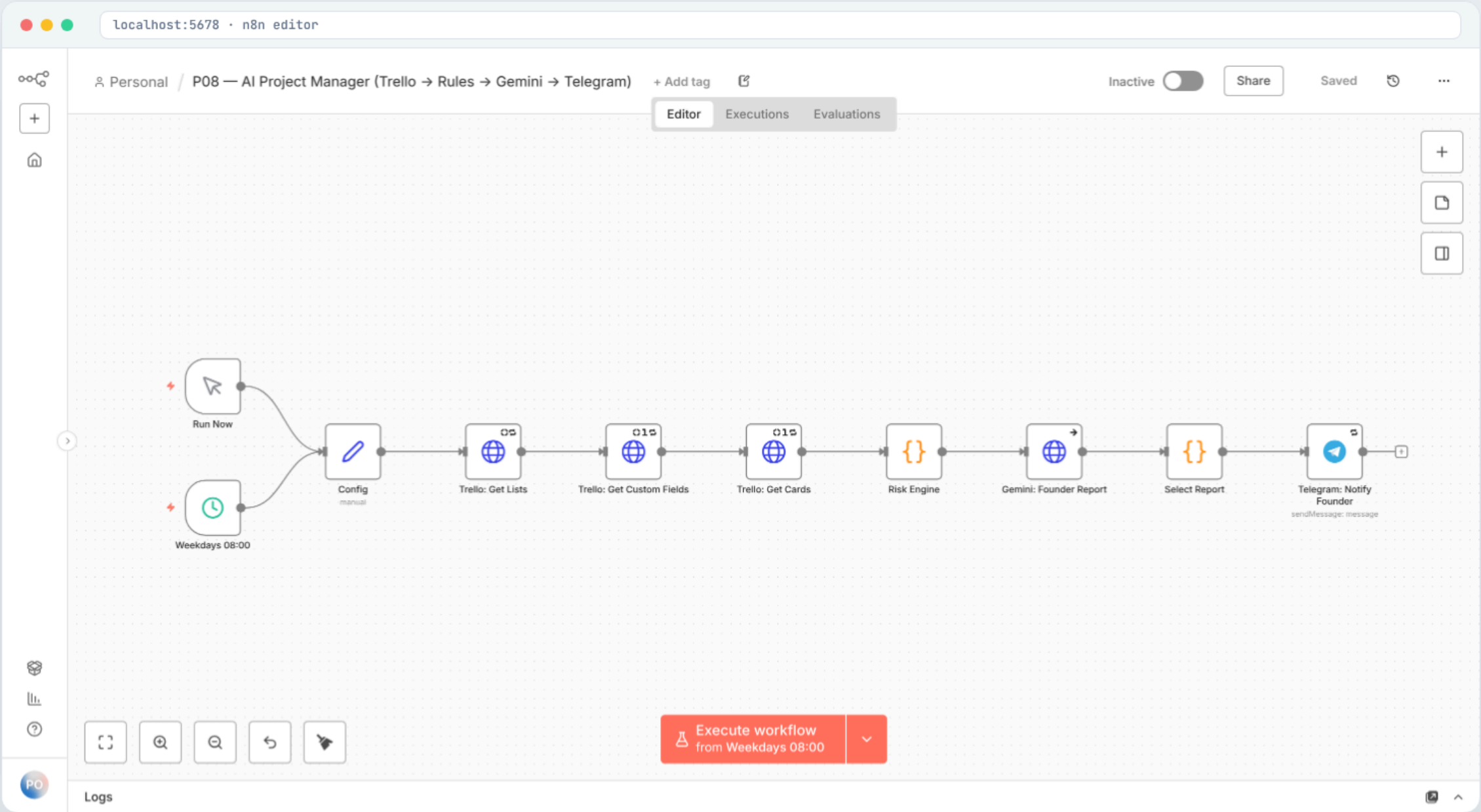
Fixed precedence: OVERDUE > BLOCKED > AT RISK > ON TRACK. Same input always gives the same answer. Thresholds are configurable.

SIGNAL	RULE	SEVERITY
● OVERDUE	Deadline before today and card not in Done	critical
● BLOCKED	Blocker field filled, or card in the Blocked list	critical
● DUE SOON, NOT STARTED	Deadline within 7 days while in Backlog / To Do	high
● DEPENDENCY AT RISK	Depends on a card that is OVERDUE or BLOCKED	high
● DEPENDENCY LATE / MISSING	Dependency finishes after this deadline, or name not found	high
● HOURS OVER ESTIMATE	Actual > estimated hours (high above +20%)	medium / high
● MISSING UPDATE	No card activity for more than 7 days (high above 14)	medium / high
● MISSING FIELDS	Owner, Deadline, Priority or Estimated Hours empty	medium
● OWNER REPORTED HIGH RISK	Risk Level custom field = High	medium

10 nodes, one straight line

Schedule or manual trigger → Config → three Trello API reads → Risk Engine → Gemini → Select Report → Telegram. Gemini errors do not stop the flow.

- n8n 1.123
- IMPORTABLE JSON
- RETRY ON TRELLO + TELEGRAM



Every project, ranked by severity

Output of the n8n run on 2026-10-07: 1 overdue, 1 blocked, 6 at risk, 3 on track. Each row says why.

● OVERDUE ● BLOCKED ● AT RISK ● ON TRACK

P08

Portfolio Health

As of 2026-10-07 · n8n execution 2026-10-07 22:05 UTC

TRELLO · SOURCE OF TRUTH

STATUS: RULE ENGINE

REPORT: RULES (AI FALLBACK READY)

ACTIVE CARDS

11

1 done

OVERDUE

1

deadline passed

BLOCKED

1

blocker recorded

AT RISK

6

one or more signals

ON TRACK

3

no signals

DECISIONS

2

waiting on founder

Projects by severity

12 CARDS

PROJECT	HEALTH	LIST	DEADLINE	HOURS	WHY
Client Portal v2 Sara Malik · Critical	● OVERDUE	In Progress	2026-10-02	138 / 120h	Deadline 2026-10-02 passed 5 day(s) ago 138h logged vs 120h estimated (+15%)
Payment Gateway Migration Omar Haddad · High	● BLOCKED	Blocked	2026-10-20	41 / 60h	Blocker: Waiting for bank sandbox credentials
Security Audit Remediation Khalid Rahman · Critical	● AT RISK	To Do	2026-10-12	0 / 24h	Due in 5 day(s) but still in "To Do"
Mobile App Release 3.4 Lina Farouk · High	● AT RISK	In Progress	2026-10-18	52 / 80h	Depends on "Payment Gateway Migration", which is BLOCKED
Support Chatbot Rollout Yusuf Amin · High	● AT RISK	In Progress	2026-10-30	20 / 45h	Depends on "Knowledge Base Cleanup" (due 2026-11-10), which finishes after this deadline (2026-10-30)
Data Warehouse Sync Bilal Qureshi · Medium	● AT RISK	In Progress	2026-10-25	55 / 40h	55h logged vs 40h estimated (+38%)
Customer Onboarding Emails	● AT RISK	Review	2026-10-28	9 / 16h	No update for 15 days

Founder report

RULE-BASED

Portfolio health — 2026-10-07

11 active: 1 overdue, 1 blocked, 6 at risk, 3 on track. 1 done.

Needs action now:

- Client Portal v2 [OVERDUE] (Sara Malik): Deadline 2026-10-02 passed 5 day(s) ago

- Payment Gateway Migration [BLOCKED] (Omar Haddad): Blocker: Waiting for bank sandbox credentials

At risk:

- Security Audit Remediation (Khalid Rahman): Due in 5 day(s) but still in "To Do"

- Mobile App Release 3.4 (Lina Farouk): Depends on "Payment Gateway Migration", which is BLOCKED

- Support Chatbot Rollout (Yusuf Amin): Depends on "Knowledge Base Cleanup" (due 2026-11-10), which finishes after this deadline (2026-10-30)

- Data Warehouse Sync (Bilal Qureshi): 55h logged vs 40h estimated (+38%)

- Customer Onboarding Emails (Hana Yusuf): No update for 15 days

TELEGRAM

The founder gets one clear message

Exact text sent by the n8n Telegram node (message accepted by the Telegram Bot API, ok = true). Gemini was unavailable in this run, so the guardrail used the rule-based report.

● DELIVERED ● AI FALLBACK PATH

P08 Project Health — 2026-10-07

● Overdue 1 ● Blocked 1 ● At risk 6 ● On track 3

Status decided by rules · report by rule engine (AI fallback)

Portfolio health — 2026-10-07

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- Security Audit Remediation (Khalid Rahman): Due in 5 day(s) but still in "To Do"
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- Data Warehouse Sync (Bilal Qureshi): 55h logged vs 40h estimated (+38%)
- Customer Onboarding Emails (Hana Yusuf): No update for 15 days
- Internal Analytics Dashboard (no owner): Missing: Owner

Decisions required from you:

- Client Portal v2: Approve a 1-week scope cut or one extra developer
- Security Audit Remediation: Approve the external pen-test vendor

On track: CRM Data Import, Knowledge Base Cleanup, Marketing Website Refresh

22:05 UTC

22 of 22 automated tests pass

Node.js built-in test runner on the engine, plus three real n8n executions covering each report path.

● 22 PASS ● 0 FAIL

- ✓ healthy card is ON TRACK with no flags
- ✓ past deadline ⇒ OVERDUE
- ✓ deadline today is not overdue
- ✓ blocker text or Blocked list ⇒ BLOCKED
- ✓ OVERDUE takes precedence over BLOCKED
- ✓ hours over estimate: medium at <=20%, high above
- ✓ missing update: >7 days medium, >14 days high, missing date flagged
- ✓ missing required fields are listed
- ✓ due soon while not started ⇒ AT RISK
- ✓ dependency on a BLOCKED card ⇒ DEPENDENCY_AT_RISK
- ✓ dependency finishing after this deadline ⇒ DEPENDENCY_LATE
- ✓ unknown, self and completed dependencies
- ✓ Done cards are never flagged and excluded from active counts
- ✓ invalid inputs are rejected or tolerated safely
- ✓ config overrides thresholds
- ✓ evaluation is deterministic
- ✓ Trello fixture: normalization reads every custom field type
- ✓ Trello fixture: expected health for every card
- ✓ closed (archived) cards are ignored
- ✓ AI guardrail: accepts faithful text, rejects omissions and contradictions
- ✓ AI prompt carries facts and forbids re-judging status
- ✓ Telegram message escapes HTML and stays within 4096 chars

Gemini unavailable

Rule-based report sent to Telegram

● PASS

Gemini answer is faithful

AI report selected

● PASS

Gemini contradicts a status

AI report rejected, rule report used

● PASS

Real Trello board

Not run in this environment

● PENDING

Real Gemini key

Not run in this environment

● PENDING

WHAT THIS GIVES A BUSINESS

A daily answer to “what needs me today?”

- ✓ Health decided by clear, testable rules
- ✓ Rule-based report if the AI is down or wrong
- ✓ Works on an existing Trello board with custom fields
- ✓ Founder report on Telegram every weekday
- ✓ Credentials kept inside n8n, never in code
- ✓ Rules and thresholds can be changed per team